

Optoelectronic resistive random access memory for neuromorphic vision sensors

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Neuromorphic visual systems have considerable potential to emulate basic functions of the human visual system even beyond the visible light region. However, the complex circuitry of artificial visual systems based on conventional image sensors, memory and processing units presents serious challenges in terms of device integration and power consumption. Here we show simple two-terminal optoelectronic resistive random access memory (ORRAM) synaptic devices for an efficient neuromorphic visual system that exhibit non-volatile optical resistive switching and light-tunable synaptic behaviours. The ORRAM arrays enable image sensing and memory functions as well as neuromorphic visual pre-processing with an improved processing efficiency and image recognition rate in the subsequent processing tasks. The proof-of-concept device provides the potential to simplify the circuitry of a neuromorphic visual system and contribute to the development of applications in edge computing and the internet of things.

The human visual system enables the recognition of various objects and visual information sensing in a complicated environment, which has inspired the development of biomimicry visual systems through electronic devices for future artificial vision^{1–3}. An artificial visual system usually consists of photoreceptors to perceive the visual inputs as digital images, a memory unit to store the visual information and a processing unit to conduct complex image-processing tasks, such as pattern recognition and object detection^{4–8}. For current artificial visual systems, state-of-the-art image sensors can continuously detect the images in real time, but generate a large amount of redundant data compared with the human visual system; this data occupies a large amount of storage space and causes high power consumption. In contrast, the sensory neurons in the retina in the human visual system not only can detect light stimuli, but also can perform first-stage image processing before the more complex visual signal processing in the visual cortex of the human brain^{9–11}. Additionally, the processing and memory units in the existing neuromorphic visual systems cannot respond directly to optical stimuli, and require image sensor arrays to convert optical signals into electrical ones and pass these to neuromorphic chips for further signal processing^{12–15}. There is a high demand to develop multifunctional electronic devices that can integrate sensing, memory and processing functions for a more efficient artificial visual system.

Towards the future development of artificial vision, optoelectronic resistive random access memory (ORRAM) and optoelectronic synaptic devices have become promising candidates, as they not only directly respond to the optical stimuli, but also allow temporary memory and real-time processing of visual information and sensory data^{16–19}. Unlike conventional image sensors that compute a linear response of light intensity (image brightness) at each pixel, ORRAM synaptic devices sense information in ways like neural signals with light-tunable and time-dependent plasticity, which enables the functions of first-stage image processing, such as contrast enhancement and noise reduction.

In this work, we designed an ORRAM synaptic device with a simple two-terminal structure of Pd/MoO_x/ITO (ITO, indium tin oxide) that exhibits ultraviolet (UV) light sensing, optically triggered non-volatile and volatile resistance switching and light-tunable synaptic behaviours. Through ORRAM arrays, we experimentally demonstrated image sensing, image memorization and real-time preprocessing functions, for example, image contrast enhancement. Further simulations of neuromorphic visual systems proved that image preprocessing (background noise reduction) at the front end can effectively improve the image quality, and increase the processing efficiency and the accuracy of subsequent image recognition.

Non-volatile switching characteristics

Optoelectronic memories can be programmed by optical stimulations and read out by electrical operations. Figure 1a illustrates the schematic structure of our designed two-terminal ORRAM with a structure of Pd/MoO_x/ITO. The inset in Fig. 1a is a cross-section scanning electron microscopy (SEM) image of the Pd/MoO_x/ITO ORRAM. The X-ray diffraction pattern and absorption spectra of the as-deposited MoO_x are shown in Supplementary Fig. 1a,b, respectively. We performed the electrical characterization by grounding the ITO electrode and sweeping the voltage on the bottom Pd electrode. The device was kept at a high resistance state (HRS) even with a large voltage window from –8 to 8 V (Supplementary Note 1 and Supplementary Fig. 1c). In sharp contrast, the device was switched to a low resistance state (LRS) under light illumination by an UV laser with a wavelength of 365 nm and a power density of 150 mW cm^{–2}. After removal of the light illumination, the device still preserved the LRS, as shown in Fig. 1b (blue line). A reset process (Fig. 1b, red line) occurred at a voltage of –2.13 V when a negative voltage sweep was applied. Figure 1c presents the results of the pulse-switching characteristics of the optical set and electrical reset. The set process was operated by an optical pulse with a power intensity of 150 mW cm^{–2} and a pulse width of 600 ms, and the reset process was initiated by an electrical pulse

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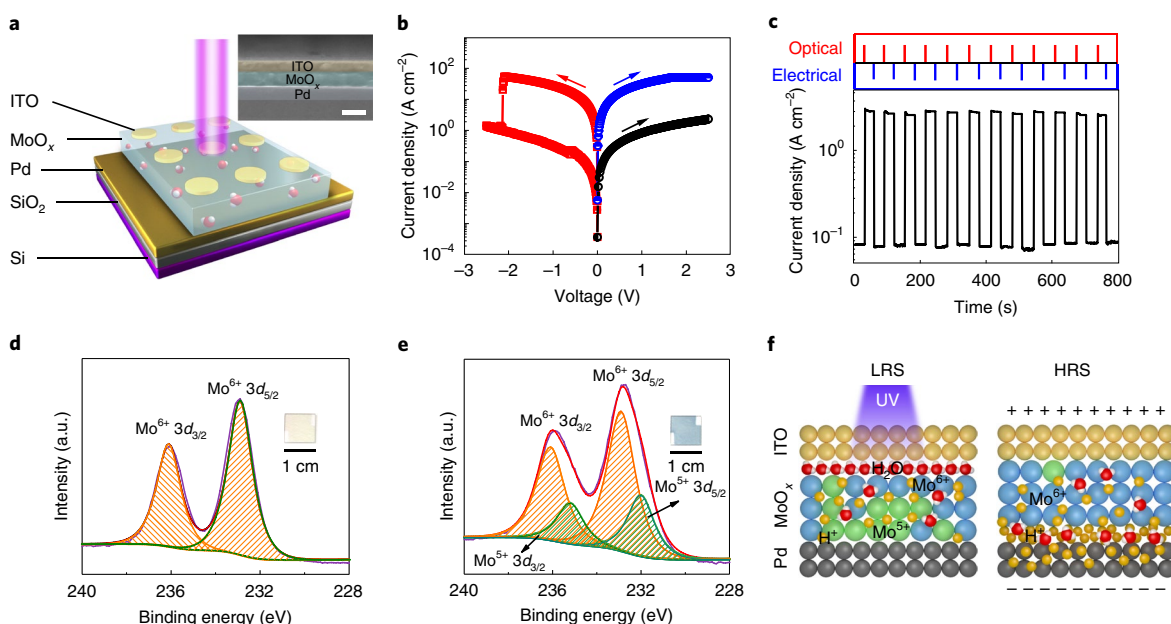


Fig. 1 | Non-volatile switching characteristics of ORRAM. **a**, Schematic structure of the MoO_x ORRAM. Pd and ITO are the bottom and top electrodes, respectively, and MoO_x is sandwiched between the bottom and top electrodes. The ITO electrode is grounded and the voltage is applied to the Pd electrode. Inset, cross-section SEM image. Scale bar, 100 nm. **b**, Optical set and electrical reset in a d.c. sweeping mode. The black and blue lines, respectively, show the voltage sweeping before and after the removal of UV illumination. The red line shows the voltage sweeping for the electrical reset process. **c**, Pulse-switching characteristics. An optical pulse (365 nm UV light) with a power intensity of 150 mW cm⁻² and a pulse width of 600 ms was used for the set process, and the reset process was initiated by an electrical pulse (−4.5 V, 100 ms). **d,e**, Narrow scans of the Mo 3d peak for the MoO_x layer before (**d**) and after (**e**) UV illumination. Insets, the photos of the MoO_x thin film prepared on a quartz substrate before (**d**) and after (**e**) UV illumination clearly show the colour transition associated with resistance change. **f**, Proposed switching mechanism in the MoO_x ORRAM. The resistance state transition is associated with the valence state change of Mo. Mo⁵⁺ and Mo⁶⁺ are represented by green and blue balls, respectively. a.u., arbitrary units.

(−4.5 V, 100 ms). The ORRAMs can be cyclically switched between the HRS and LRS and delivers an on/off ratio of ~40 at a read voltage of 0.1 V. Supplementary Fig. 1d,e show the retention test after the pulse switching and cumulative probability plot, respectively, which demonstrate typical non-volatile characteristics with small device-to-device variation. The MoO_x ORRAM also exhibits area-dependent switching characteristics (Supplementary Fig. 1f), which suggests the switching mechanism of the device is homogeneous rather than filamentary^{20,21}. The top and bottom electrodes of the ORRAM device can potentially be scaled to the nanoscale, exhibiting comparable switching characteristics (Supplementary Note 2 and Supplementary Fig. 2).

To understand the origin of the change of resistance states, we carried out characterizations using X-ray photoelectron spectroscopy (Supplementary Fig. 3). Figure 1d,e shows the corresponding narrow scans of the Mo 3d core level spectra of MoO_x before and after the UV illumination. At the initial state before the UV illumination, the MoO_x film showed Mo 3d_{3/2} (236.0 eV) and Mo 3d_{5/2} (232.9 eV) peaks, which fit Mo⁶⁺ states well. After the UV illumination, Mo 3d peaks were fitted and dissolved into four peaks, Mo⁶⁺ 3d_{3/2} (236.0 eV), Mo⁶⁺ 3d_{5/2} (232.9 eV), Mo⁵⁺ 3d_{3/2} (234.9 eV) and Mo⁵⁺ 3d_{5/2} (231.8 eV), which suggests the formation of Mo⁵⁺ ions induced by UV illumination. The ratio of Mo⁵⁺/(Mo⁵⁺ + Mo⁶⁺) in the film is extracted as 0.31. The change of valence state of Mo ions has been reported to result from the incorporation of protons that are produced from water dissociation due to the UV illumination^{22–25}. We also performed a first-principle calculation using density functional theory to reveal the electronic structure of α-MoO₃ before and after the intercalation of protons (Supplementary Note 3 and Supplementary Fig. 4). The results suggest the transition from semiconducting to metallic characteristics. The Fermi level shifted

towards the conduction band after the intercalation of protons, which is also consistent with previous theoretical calculations^{26,27} and experimental results^{24,28,29}. The conductive H_yMoO_x formed a percolating conduction network in the MoO_x thin film, which gave rise to the LRS. The transition of valence state is also associated with the colour change in the MoO_x film, which is evidenced by the insets of Fig. 1d (transparent) and 1e (blue). We used lateral devices with different metal electrodes (Pd, Au, Mo, Ti and Al) to observe the relationship between colour change and resistance states under different UV illumination conditions (Supplementary Note 4 and Supplementary Fig. 5).

Based on the non-volatile switching characteristics of the ORRAM, we propose a valence change switching mechanism (Fig. 1f). As the MoO_x thin film absorbs UV light from the top ITO electrode, electrons and holes are generated in the MoO_x film. The photogenerated electrons are excited to the conduction band of MoO_x. The protons (H⁺) are produced by the reaction between photogenerated holes and the absorbed water molecules in the MoO_x thin film^{22,23,25,30}. The photogenerated electrons and the protons can lead to a change of valence state of the Mo ions (from 6⁺ to 5⁺) and the formation of hydrogen molybdenum bronze (H_yMoO_x), which results in the transition of resistance states. During the reset processes, the electric field drives the protons to drift towards the Pd electrodes. As the protons are extracted from the MoO_x layer, the valence state of the Mo ions changes from 5⁺ back to 6⁺, which leads to the transition from LRS to HRS. The Supplementary Information gives a detailed explanation of the non-volatile switching mechanism and supplementary experiments, which include moisture-dependent switching (Supplementary Note 5 and Supplementary Fig. 6), electrode-material-dependent switching (Supplementary Fig. 7), temperature-dependent switching (Supplementary Fig. 8), a band

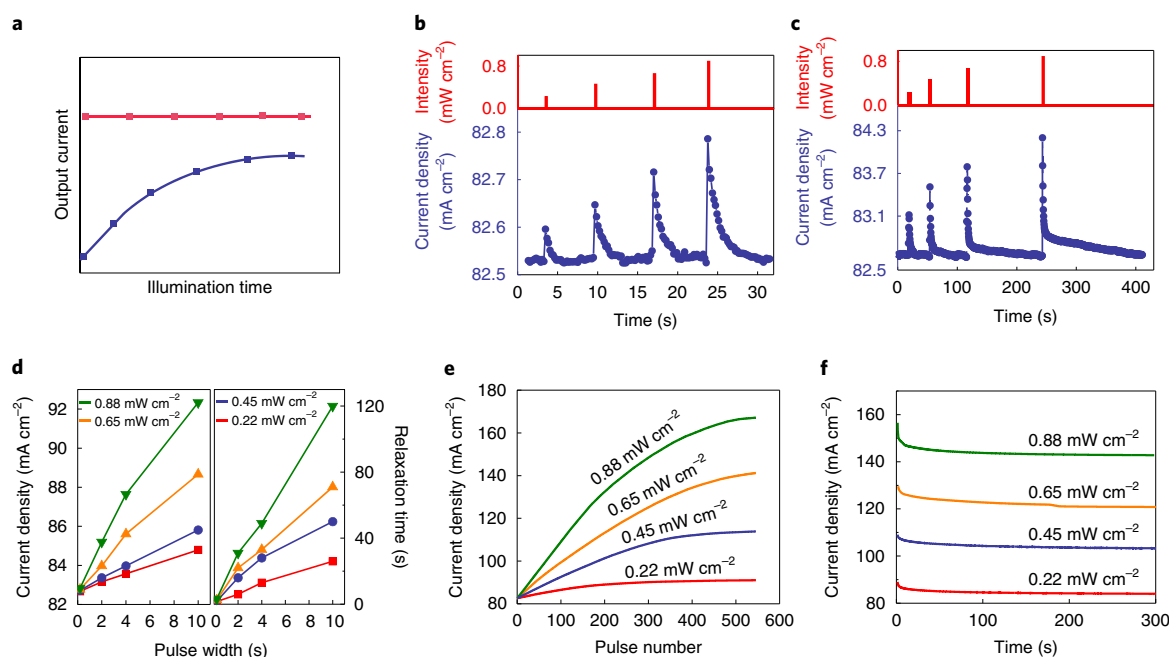


Fig. 2 | Light-tunable synaptic characteristics. **a**, Comparisons of output current versus the light-illumination time of a conventional image sensor (red) and ORRAM synaptic device (blue). The conventional image sensor exhibits no time-dependent response, whereas the ORRAM synaptic device exhibits an increased output current with illumination time ($I = K(t) + C$, in which K is a function related to time and C is a constant). **b**, Light-intensity-dependent (0.22, 0.45, 0.65 and 0.88 mW cm⁻²) STP with a pulse width of 200 ms. **c**, Light-intensity-dependent STP with a pulse width of 2 s. **d**, Plots of spike current and relaxation time as a function of pulse width at different light intensities. **e**, Long-term potentiation with the pulse number increased to 500 (pulse width, 200 ms; pulse interval, 200 ms). **f**, Transition from STP to LTP after 300 pulse stimulations at different light intensities. The retention test was monitored with a small readout voltage of 0.1 V.

diagram of the as-deposited MoO_x thin film (Supplementary Fig. 9) and thickness-dependent electrical measurements (Supplementary Note 6, Supplementary Fig. 10 and Supplementary Table 1).

Light-dosage tunable synaptic characteristics

The ORRAM can be programmed by optical stimulation and output light-dosage-dependent and history-dependent resistance states, which allow us to mimic the basic features of synaptic plasticity in emulating the learning and memory functions of the human brain, which include short-term plasticity (STP) and long-term plasticity (LTP). Figure 2a schematically illustrates the output current as a function of the illumination time of a conventional image sensor and the ORRAM synaptic device. Both devices exhibit light-intensity-dependent characteristics. However, the conventional image sensors can only respond to light intensity. In contrast, the ORRAM device exhibits a time-dependent output current, which allows further complicated processing functions. To examine the integrated light-sensing capabilities and synaptic functions, we investigated the effects of light dosage (light intensity and illumination time) on the synaptic behaviours.

Figure 2b shows the STP under optical stimuli (pulse width of 200 ms) with different light intensities. By employing relatively weak optical stimuli, the ORRAM is switched to a more conductive state, generates a current spike, and then decays to the initial conductance state. The current decay is possibly a result of the weak interaction between protons and MoO_x under a weak optical stimulus. A high light intensity gave rise to an enhanced synaptic strength, which includes a large spike current and long relaxation time. This characteristic can be understood as a result of the large amount of photogenerated charge carriers and protons generated per unit time under optical illumination with a high intensity, which leads to a high concentration of protons and a high reaction rate for generating Mo⁵⁺.

Complicated image processing requires a response to both light intensity and light dosage. We further investigated the effects of light illumination time and the total light dosage on the synaptic behaviours. As the pulse width of the optical stimulus was further raised to 2 s, the synaptic strengths increased significantly (Fig. 2c). Figure 2d shows the spike current and relaxation time as a function of pulse width under four different light intensities. The relaxation time was extracted as the time taken for the current to drop to 30%. The rates of the spike current and relaxation time increase with the light intensity as a function of pulse width or pulse number, owing to the improved reaction rate and increased number of protons and Mo⁵⁺. In biological systems, the transition from STP to LTP can be achieved by employing a repeated pulse stimulation that persistently increases the synaptic strength. To mimic the transition from STP to LTP, we conducted experiments by applying 500 identical pulses repeated successively (Fig. 2e). The current increases steadily with the stimulation pulse number and exhibits light-sensing behaviour.

We also examined the retention characteristics after 300 pulses for 300 s (Fig. 2f). The transition from STP to LTP can be clearly observed with a largely increased retention time after a 300-pulse stimulation. The increased pulse number can induce an increased number of photogenerated charge carriers, which can further accelerate the water dissociation reaction and transition to Mo⁵⁺ and increase the probability of a Mo valence state change from 6⁺ to 5⁺. Additionally, a long exposure time to the light can induce defect states, which reduce the charge carrier recombination, result in a more stable proton and Mo⁵⁺, and improve the resistance retention properties^{31–33}. Experimentally, spectroscopic studies proved that a long light exposure gave increased defect states (Supplementary Note 7 and Supplementary Fig. 11). In addition to the tunable synaptic behaviour triggered by the UV light (365 nm), the MoO_x thin film can also respond to blue light through tuning the stoichiometry

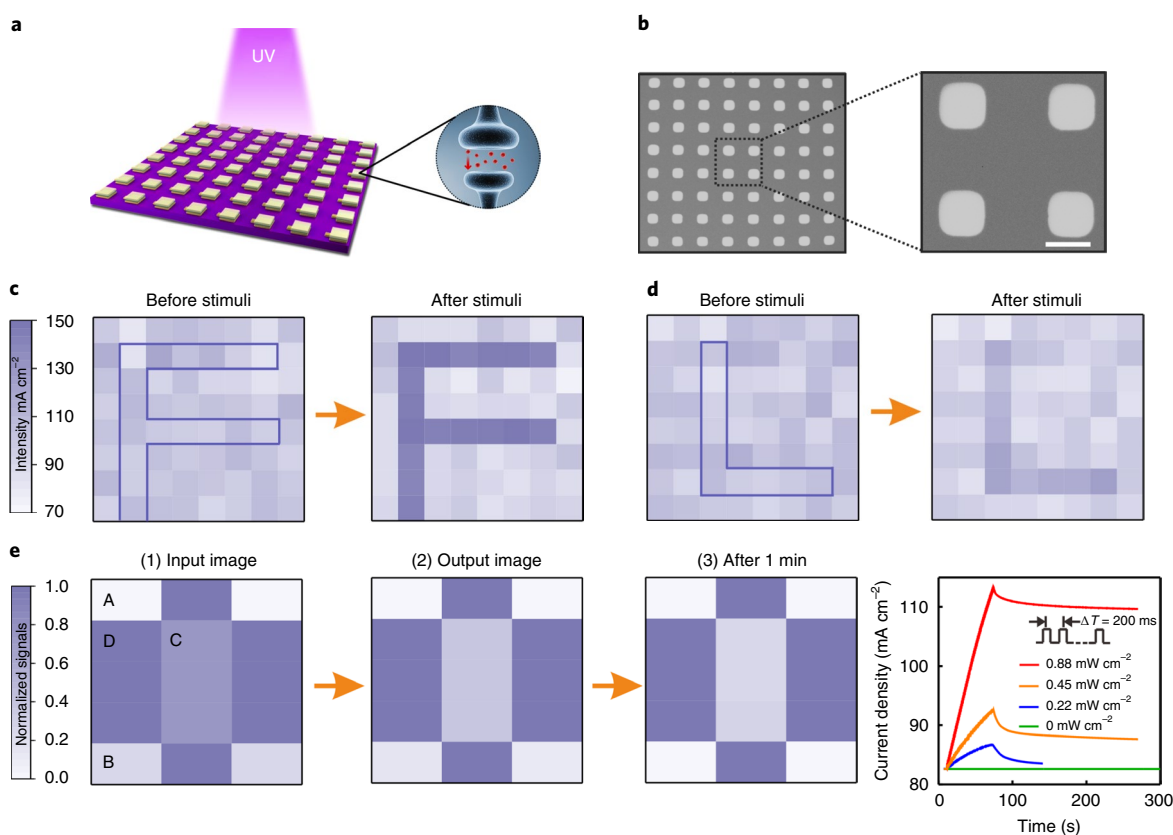


Fig. 3 | Image memorization and preprocessing. **a**, Schematic structure of an 8 × 8 ORRAM array. **b**, SEM image of an 8 × 8 ORRAM array. Scale bar, 200 μm. **c,d**, Illustrations of the image memory of the letter F (**c**) and the letter L (**d**). The letter F was stimulated with a light intensity of 0.88 mW cm⁻² and the letter L with a lower light intensity of 0.45 mW cm⁻² (pulse width, 200 ms; pulse interval, 200 ms). **e**, An illustration of the image contrast enhancement after the training processes (1)–(3) and the current response (right panel) during and after the training processes. The training process was conducted through an input image repeated 100 times with a frequency of 2.5 Hz (pulse width, 200 ms; pulse interval, 200 ms). The four grey scales in the input image correspond to four light intensities (0 (A), 0.22 (B), 0.45 (C) and 0.88 (D) mW cm⁻²). The signals in the output image correspond to the current after the training process. The input signals (light intensity) and output signals (current density) are normalized.

and bandgap. Detailed information is given in Supplementary Note 8 and Supplementary Figs. 12–15.

Image memorization and preprocessing

A conventional artificial photoreceptor consists of an image sensor based on Si CMOS (complementary metal–oxide semiconductor) technology and a field-programmable gate array to convert an image into spike signals. Different from conventional photodetectors or image sensors, ORRAMs can output light-dosage-dependent resistance states, which enables an integral image, weak signal accumulation, spectrum analysis and other complicated image-processing functions³⁴. We construct ORRAM arrays (Fig. 3) to demonstrate basic functions of image sensing, image memory and image contrast enhancement. Figures 3a,b illustrates the schematic structure and SEM images of an 8 × 8 ORRAM array.

The image memorizations of the letters F and L with illuminated intensities of 0.88 and 0.45 mW cm⁻² are illustrated in Fig. 3c,d, respectively. For this illustration, one memory cell corresponds to one image pixel. The readout circuit is illustrated and described in Supplementary Note 9 and Supplementary Fig. 16. Optical masks with the shapes of F and L were used during the UV light illumination. Two types of stimuli were employed: (1) the stimuli for letter F were 300 optical pulses with an intensity of 0.88 mW cm⁻² and (2) the stimuli of letter L were 300 optical pulses with an intensity of 0.45 mW cm⁻². We recorded the images before the light stimuli and three minutes after the removal of the light stimuli. Before applying

the stimuli, the devices were at HRS. After the light stimuli, the conductance of the memory devices increased and the memory effects of the letters F and L were observed. The letter F exhibited a stronger memory effect than the letter L owing to the stronger light stimuli. The above processes suggest the capability of image memory through ORRAM array devices.

Image preprocessing is a main characteristic of the human retina, which plays a significant role in improving the quality of the sensory data and further increasing the processing efficiency and accuracy of subsequent tasks, such as image recognition and classification. For image preprocessing, contrast enhancement and noise reduction are essential to highlight the main features of an image. According to the light-tunable plasticity above, the spike current and retention time of ORRAM devices increase faster under a higher light intensity. Correspondingly, the pixel with a higher brightness shows a more enhanced accumulation effect. After repeated training, the difference between the grey scale of the pixels (normalized) is enlarged over the input images, and thus outputs an image with an enhanced contrast.

Based on this aspect, we employed a 3 × 5 ORRAM array to demonstrate the contrast enhancement function of an image with four grey scales. During the training processes, an image was repeatedly input into the ORRAM array 100 times with a frequency of 2.5 Hz. Image (1) in Fig. 3e is an example of an input image in which four grey scales in the input image correspond to four light intensities of 0 mW cm⁻² (pixel A), 0.22 mW cm⁻² (pixel B), 0.45 mW cm⁻² (pixel C) and 0.88 mW cm⁻² (pixel D). The intensities are normalized

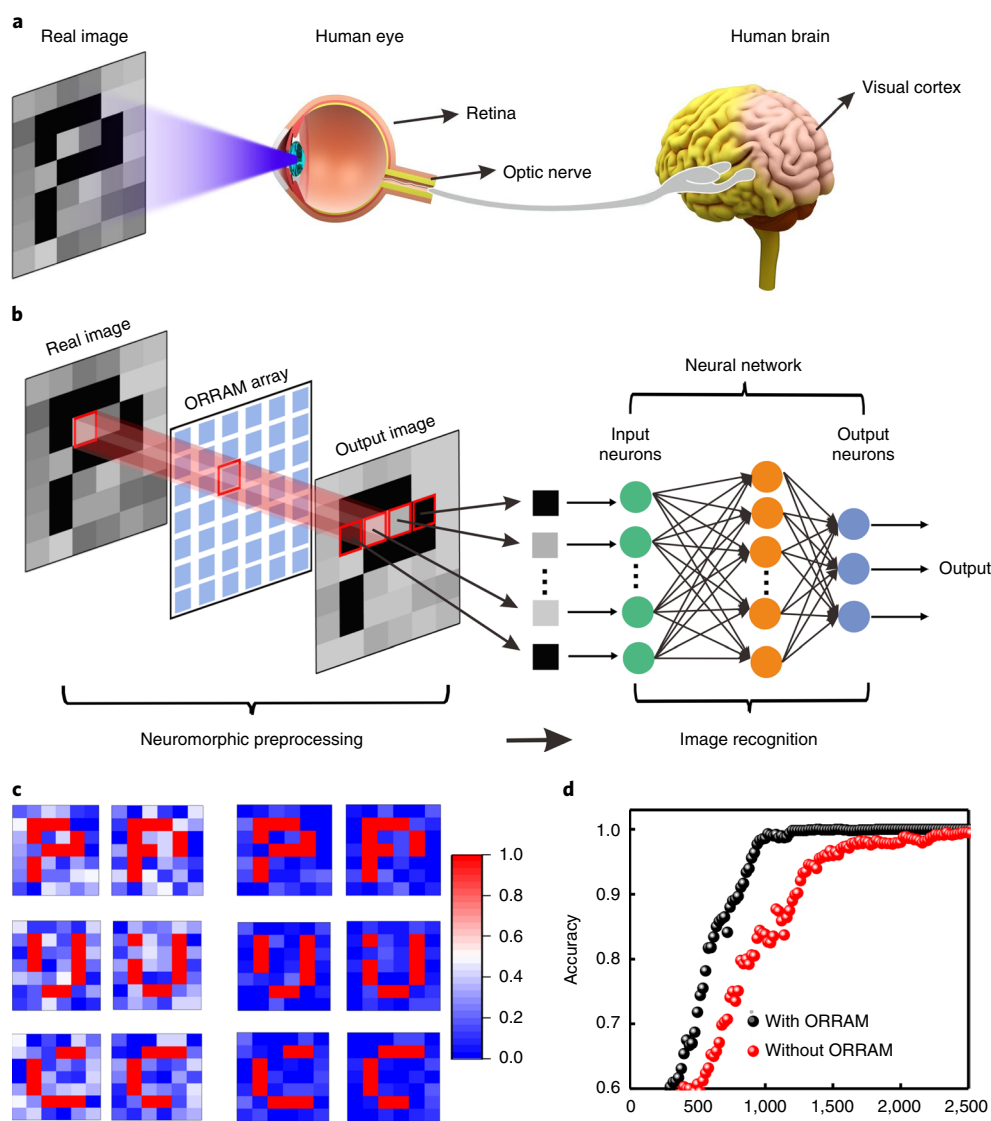


Fig. 4 | Simulations of image recognition in a neuromorphic visual system with ORRAM. **a**, Schematics of the human visual system. **b**, Illustration of an artificial neuromorphic visual system based on the ORRAM devices for image preprocessing and an artificial neural network for image recognition. **c**, Examples of images before (left columns) and after (right columns) ORRAM-based pre-processing. **d**, Comparisons of the image recognition rate with and without ORRAM-based image preprocessing.

to 0–1 such that the input optical signals of the four pixels A, B, C and D are 0, 0.25, 0.5 and 1, respectively. After repeated inputs of the image 100 times, the output currents respond according to the different light intensities, as shown in Fig. 3e (right panel). After normalization of the currents from 0 to 1, the output currents of pixels A, B, C and D were 0, 0.13, 0.39 and 1, respectively (the weakest signals are normalized to 0, and the strongest signals are normalized to 1 for both input and output images). An output image can be reshaped according to the output current values through the ORRAM devices. The differences in current between pixels A, B, C and D are enlarged compared with those in the input image, which results in an image with an enhanced contrast (image (2) in Fig. 3e). After one minute, the currents that correspond to pixels with a lower intensity decay faster, whereas the currents from pixels with a higher intensity decay slower. The corresponding values of pixels A, B, C and D after the normalization are 0, 0.01, 0.27 and 1, respectively. In this case, the contrast of the output image is further enhanced (image (3) in Fig. 3e). The increase in the rate of the current in the

ORRAM devices with a higher light intensity is much higher than that with a lower light intensity. In this way, the main feature in one image is highlighted, and the image contrast is enhanced.

Simulations of image recognition with preprocessing

Compared with a conventional system based on von Neumann computing architecture and Si CMOS technologies, neuromorphic visual systems based on artificial neural network have delivered promising results in building artificial visual systems, which involve efficiently processing visual information, like the visual cortex in human brain, with reductions in the amount of hardware and power consumption.

As image preprocessing of contrast enhancement can effectively highlight the image body features, we performed simulations to preprocess more complex images. We constructed two components to mimic the human visual systems—preprocessing based on ORRAM arrays and image recognition based on an artificial neural network. Figure 4a depicts the schematics of the human visual system.

The visual information is first detected and extracted through the retina in the human eye, and the information is further passed through optic nerves and processed in the visual cortex. In the neuromorphic visual system, ORRAM arrays are employed to emulate the sensing and preprocessing functions in the human retina. The preprocessed images are transported to a three-layer artificial neural network to complete the image training and recognition functions, as shown in Fig. 4b. A neural network with an input layer (42), hidden layer (20) and output layer (3) was established in the simulation for image recognition. The detailed architecture of the neural network in this work is described in Supplementary Note 10 and Supplementary Fig. 17.

The artificial visual system in Fig. 4b contains image sensing, preprocessing and recognition. An image database that consisted of images of the letters P, U and C with 6×7 pixels was adopted for the artificial neuromorphic visual systems. Two image examples of each letter from the database (1,000 images) were extracted for demonstration. All the signals (light intensity) in the image were normalized. The background noise signals in each image were randomly generated in a range of 0–0.5. The fitting line of the experimental data (current density versus light intensity) for the simulation is shown in Supplementary Fig. 18. Detailed information of the database is given in Supplementary Note 10 and Supplementary Fig. 19. The input and output images after the preprocessing are compared in Fig. 4c. After the preprocessing through ORRAMs, the body features of the letters were highlighted and the background noise signals were smoothed. The preprocessed image database was then input into the artificial network to conduct the image training and recognition. Figure 4d illustrates the recognition rate of visual systems with and without ORRAM, and shows obvious improvements in the recognition rate and efficiency with the use of ORRAMs for the image preprocessing. Without the image preprocessing by ORRAMs, the recognition rate reaches 0.980 after 2,000 training epochs, whereas it only requires 1,000 training epochs to reach a recognition rate of 0.986 with preprocessing through ORRAM. The recognition rates of different running cycles are shown in Supplementary Fig. 20, which suggests the repeatability of the results. The image-recognition efficiency, which includes processing speed and energy consumption, proved to be largely improved (a 41.5% improvement to reach a recognition rate of 0.99) with the image preprocessing at the front end. Detailed calculations and comparisons are listed in Supplementary Note 10 and Supplementary Table 2. These results suggest that image preprocessing can be completed through the ORRAM devices, which also further improves the image-processing capabilities.

Conclusions

In this work, we designed and constructed a MoO_x ORRAM synaptic device that can respond to optical stimuli directly. The device can store the optical information and execute light-tunable synaptic functions. The UV-light-tunable synaptic plasticity of the ORRAM array allows us to perform a first-stage image processing, such as image contrast enhancement and noise reduction. By performing image preprocessing with ORRAMs, the processing efficiencies and accuracy of subsequent processing tasks are effectively improved. The ORRAM synaptic devices exhibited great potential in the application of a neuromorphic visual system by simplifying the circuitry and reducing the power consumption, which helps to efficiently process an overwhelming amount of dynamic visual information from the surrounding environment.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, statements of code and data availability and associated accession codes are available at <https://doi.org/10.1038/s41565-019-0501-3>.

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Author contributions

Y.C. and F.Z. conceived and designed the project. Y.C. supervised the project. F.Z., T.H.C., J.W. and Z.L. performed the experiments, including both fabrication and characterization. Z.Z., F.Z., J.C., S.Y. and J.K. performed the simulations. J.C. performed the density functional theory calculations. F.Z., N.Z., Y.C., S.Y., J.K. and H.-S.P.W. analysed the data. F.Z. and Y.C. co-wrote the paper. All the authors discussed the results and commented on the manuscript.

Additional information

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Methods

Device fabrication of MoO_x optoelectronic memory and array device. The MoO_x optoelectronic memories were fabricated in a configuration of Pd/MoO_x/ITO, in which a Pd layer 100 nm thick (the bottom electrode) was first deposited on the SiO₂/Si substrate by electron-beam evaporation with a base pressure of 8×10^{-7} Torr and a deposition rate of $0.5 \text{ \AA s}^{-1}$. Then a MoO₃ layer of thickness 80 nm was deposited by thermal evaporation at an evaporation rate of $1.5 \text{ \AA s}^{-1}$. The base pressure was 5×10^{-6} Torr with an evaporation pressure of 9×10^{-5} Torr. Before the deposition of the top ITO electrode, a dot-shaped shadow mask was attached on top of the MoO_x to define the effective area of the top electrode. The transparent ITO was deposited as the top electrode by d.c. sputtering. The effective device area was defined by a dot-shaped shadow mask with a diameter of 250 μm . The sputter base pressure was controlled at 8×10^{-7} Torr and the deposition pressure was controlled at 3 mTorr. The deposition power was 50 W, which corresponded to a deposition rate of $0.3 \text{ \AA s}^{-1}$. For the array devices, the three layers were all patterned by shadow masks with an effective device area of $200 \mu\text{m} \times 200 \mu\text{m}$.

Material characterization of the MoO_x layer. SEM was performed with a Hitachi S-4800 field emission scanning electron microscope. X-ray diffraction patterns of the MoO_x film was determined by a Rigaku SmartLab X-ray diffractometer with a 2θ range from 10 to 70° in steps of 0.01°. The absorption spectra of the MoO_x film on quartz were examined using a UV-2550 Shimadzu UV-visible spectrophotometer. X-ray photoelectron spectroscopy was carried out in a vacuum chamber

($<10^{-5}$ Torr). A monochromatic Al K α (photon energy, 1,486.6 eV) source was used as the excitation source for the X-ray photoelectron spectroscopy. A pass energy of 80 or 40 eV was employed for the wide or core-level narrow scan, respectively. The core-level signals were recorded at a photoelectron take-off angle of 90°. All the binding energies were calibrated with reference to the C 1s peak at 284.6 eV.

Device characterization. All the electrical characterizations were conducted using a probe station and a Keithley 4200. ITO electrodes were grounded and the voltage was applied to the Pd electrode. An UV laser and light-emitting diode with a wavelength of 365 nm were used in the experiment for the optical switching. The readout voltage used in the test was 0.1 V. To demonstrate the image memorization, we used an optical mask. The pixels in the letters were illuminated simultaneously, and the pixel currents were read sequentially. To demonstrate the image contrast enhancement, each pixel was illuminated and measured one by one.

Data availability

The data that support the plots within this paper and other findings of this study are available from the corresponding author upon reasonable request.

Code availability

The simulation codes used for this study are available from the corresponding author upon reasonable request.